

Weekly Notice 10

Exercises, week 10:

1. Let $(B_t)_{t \geq 0}$ denote a standard Brownian motion.
 - (a) Determine the joint distributions of $(B_1, B_2 - B_1, B_e - B_2)^\top$ and of $(B_1, B_2 - B_1, B_e - B_1)^\top$. (Recall that $e \approx 2.718 \dots$) .
 - (b) Determine $\mathbb{E}[B_1^3 B_\pi]$.
 - (c) Determine $\mathbb{P}(B_1 < 0 < B_2 \mid \mathcal{F}_1^B)$ and $\mathbb{E}[\cos(M_1) \cdot B_2 \mid \mathcal{F}_1^B]$, where $M_1 = \sup_{u \in [0,1]} B_u$.
2. Exercise 80.
3. Exercise 93.
4. Exercise 101.
5. Exercise 104. Note that we are in continuous time so it is not enough to verify the martingale property "one-step ahead".

For those of you who do not yet know what a martingale is, here is a more detailed explanation:

 - In (1), you must show that for $0 \leq s < t$, we have $\mathbb{E}[Y_t \mid \mathcal{F}_s] = Y_s$.
 - In (2), you must show that for $0 \leq s < t$, we have $\mathbb{E}[Y_t^2 - \sigma^2 t \mid \mathcal{F}_s] = Y_s^2 - \sigma^2 s$.
6. Exercise 110. Also determine the distribution of B_{α_1+7} .